Classroom Activity: Subcategorization (Way 1: Split V into sub-categories)

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Quick recap

We split the single category **V** into smaller groups. Each group has its **own VP rule**, so a verb can only be used with the words it needs after it.

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Verb group  VP rule               Example
V_intr      VP → V_intr           Ravi slept.
V_tr        VP → V_tr NP          Priya found [a letter].
V_ditr      VP → V_ditr NP NP     Ravi gave [Priya] [a book].
V_loc       VP → V_loc NP PP      Ravi put [the book] [on the table].
V_sent      VP → V_sent S         Priya thinks [Ravi slept].
V_inf       VP → V_inf VP_inf     Ravi wants [to sleep].
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intr = intransitive (no object), tr = transitive (one object), ditr = ditransitive (two objects), loc = object + place, sent = takes a sentence, inf = takes "to + verb".

Activity 1: Write the rule for the sentence

For each sentence, find the **main verb**, write its **verb group** (V_intr, V_tr, V_ditr, V_loc, V_sent or V_inf) and write the **VP rule** it needs. The first row is done for you.

No	Sentence	Main verb	Verb group	VP rule
Ex.	*Ravi slept.*	*slept*	*V_intr*	*VP → V_intr*
1	Meena laughed.	laughed	V_intr	VP → V_intr
2	Arjun reads a novel.	reads	V_tr	VP → V_tr NP
3	The teacher gave Kavya a pen.	gave	V_ditr	VP → V_ditr NP NP
4	Priya put the book on the table.	put	V_loc	VP → V_loc NP PP
5	Ravi believes Meena won.	believes	V_sent	VP → V_sent S
6	The baby slept.	slept	V_intr	VP → V_intr
7	Kumar sent his friend a letter.	sent	V_ditr	VP → V_ditr NP NP
8	Divya ate an apple.	ate	V_tr	VP → V_tr NP
9	Arun said the bus arrived.	said	V_sent	VP → V_sent S
10	Lakshmi wants to dance.	wants	V_inf	VP → V_inf VP_inf

No	Sentence	Main verb	Verb group	VP rule
11	The students showed the principal their project.	showed	V_ditr	VP → V_ditr NP NP
12	Raj kept the keys in the drawer.	kept	V_loc	VP → V_loc NP PP

Part B: Write the complete grammar

Write all the rules needed to accept each sentence: the phrase rules **and** the lexicon. Put each verb under its correct verb group (V_intr, V_tr, ...).

1. The teacher gave Kavya a pen.

S → NP VP NP → Name | Det N

VP → V_ditr NP NP

V_ditr → gave

Name → Kavya Det → the | a

N → teacher | pen

2. Ravi believes Meena won.

S → NP VP NP → Name

VP → V_sent S

VP → V_intr

V_sent → believes V_intr → won

Name → Ravi | Meena

3. **Think:** "Ravi ate." and "Ravi ate an apple." are both correct English. Which verb group(s) should **ate** be listed under? Why is one group not enough?

“ate” should be listed under BOTH V_tr and V_intr (V_tr → ate and V_intr → ate).

“Ravi ate.” has no object, so it needs VP → V_intr; “Ravi ate an apple.” needs VP → V_tr NP.

Each group has only one VP rule, so a single group cannot allow both an object and no object.

Activity 2: Apply the rules - Accept or Reject?

Use **only** the grammar below. For each sentence, write **ACCEPT** or **REJECT**. If rejected, write the reason: which VP rule the verb must use, and what is **missing** or **extra**. If accepted, you may write the bracketed structure.

Grammar

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S  → NP VP          NP → Name | Det N          PP → P NP


VP → V_intr
VP → V_tr NP
VP → V_ditr NP NP
VP → V_loc NP PP
VP → V_sent S
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Lexicon

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V_intr → slept | arrived | laughed
V_tr   → found | ate | read
V_ditr → gave | sent
V_loc  → put
V_sent → thinks | said
Name → Ravi | Priya | Meena | Arun        Det → a | the
N    → book | letter | apple | pen | table | bag | cake
P    → on | in
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No	Sentence	Accept / Reject	Reason
Ex.	*Ravi found.*	*REJECT*	*found is V_tr, so VP → V_tr NP must be used. The NP is missing.*
1	Meena arrived.	ACCEPT	[S [NP [Name Meena]] [VP [V_intr arrived]]]
2	Arun found.	REJECT	found is V_tr, so VP → V_tr NP must be used. The NP is missing.
3	Priya gave Ravi a pen.	ACCEPT	[S [NP Priya] [VP [V_ditr gave] [NP Ravi] [NP [Det a] [N pen]]]]
4	Priya slept the book.	REJECT	slept is V_intr, so VP → V_intr must be used. The NP “the book” is extra.
5	Ravi put the book on the table.	ACCEPT	[S [NP Ravi] [VP [V_loc put] [NP [Det the] [N book]] [PP [P on] [NP [Det the] [N table]]]]]
6	Meena said the apple.	REJECT	said is V_sent, so VP → V_sent S must be used. It needs a sentence (S) after it; the NP “the apple” is not an S.
7	Arun sent Meena a letter.	ACCEPT	[S [NP Arun] [VP [V_ditr sent] [NP Meena] [NP [Det a] [N letter]]]]
8	Ravi put the book.	REJECT	put is V_loc, so VP → V_loc NP PP must be used. The PP (place phrase) is missing.
9	Meena said Arun slept.	ACCEPT	[S [NP Meena] [VP [V_sent said] [S [NP Arun] [VP [V_intr slept]]]]]
10	Priya gave a pen.	REJECT	gave is V_ditr, so VP → V_ditr NP NP must be used. The second NP is missing (only one NP, “a pen”).
11	Meena read the letter.	ACCEPT	[S [NP Meena] [VP [V_tr read] [NP [Det the] [N letter]]]]

No	Sentence	Accept / Reject	Reason
12	Arun laughed the cake.	REJECT	laughed is V_intr, so VP → V_intr must be used. The NP “the cake” is extra.
13	Ravi thinks Priya ate the cake.	ACCEPT	[S [NP Ravi] [VP [V_sent thinks] [S [NP Priya] [VP [V_tr ate] [NP [Det the] [N cake]]]]]]
14	Priya thinks Ravi found. (challenge)	REJECT	thinks is V_sent and correctly takes an S, but inside that S, found is V_tr and needs VP → V_tr NP. The NP is missing after found.
15	Ravi ate the apple on the table. (challenge)	REJECT	ate is V_tr, so VP → V_tr NP must be used. The PP “on the table” is extra: only V_loc takes a PP, and the grammar has no rule that attaches a PP to a V_tr VP or to an NP.